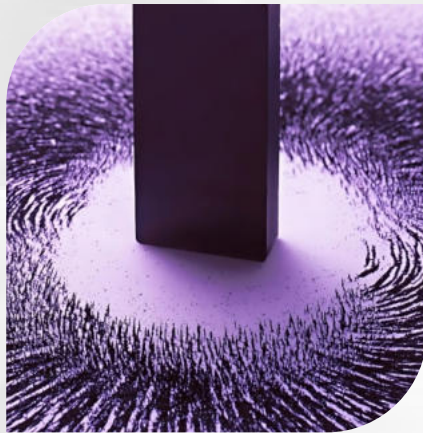
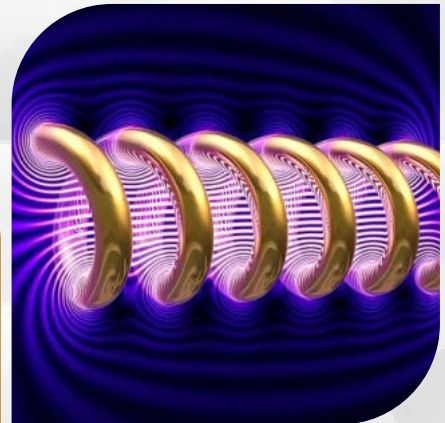


UNIT 16

ELECTROMAGNETISM



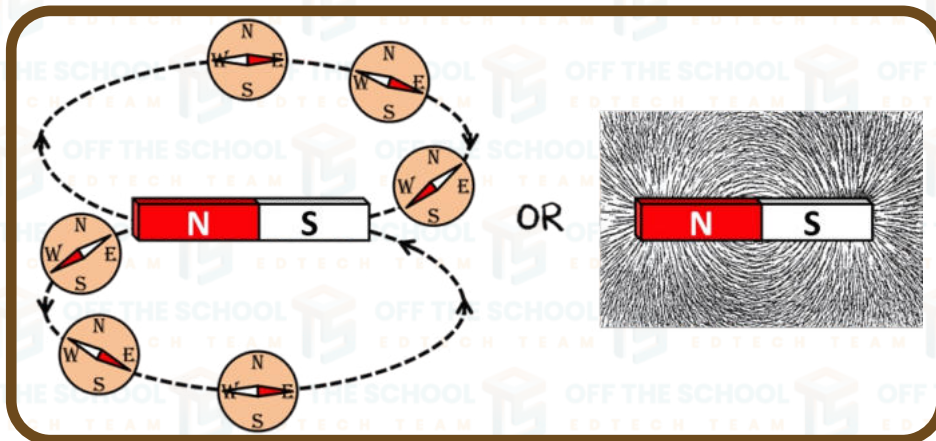
ELECTROMAGNETIC FORCE

The electromagnetic force is a fundamental interaction between charged particles, encompassing both electric attraction and repulsion as well as magnetic interactions, governed by Maxwell's equations and quantified through the concept of electromagnetic fields.

EXAMPLE

Electric Attraction: When a positively charged balloon is brought close to a negatively charged balloon, they are pulled towards each other due to the electromagnetic force.

Magnetic Interaction: A compass needle aligns with the Earth's magnetic field, demonstrating the magnetic interaction arising from the electromagnetic force.



MAGNETIC FIELD PRODUCED BY CURRENT CARRYING CONDUCTOR

A magnetic field produced by a current-carrying conductor is a region of influence around the conductor where moving electric charges generate a circular pattern of magnetic lines of force perpendicular to the direction of current flow. This magnetic field can induce magnetic interactions with nearby objects and other currents.

Now suppose a particle carrying charge q is projected with speed v into a magnetic field of magnetic induction B such that the angle between B and V is θ . The magnetic field of the charged particle interacts with the magnetic field of the magnet in which it is sent, due to which a force is produced which acts upon the particle. It is found that:

• The force F acting on the particle is directly proportional to the charge q .

• The force F acting on the particle is directly proportional to the velocity V .

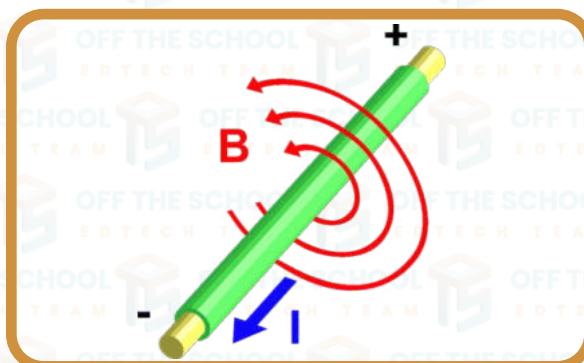
• The force F is directed perpendicular to the plane containing V and B .

Combining the above three observations, we found that:

$$F = q (V \times B)$$

So the magnitude of B is given by:

$$B = \frac{F}{qv \sin \theta} = \frac{\text{Newton}}{\text{Coulomb} \cdot \text{m/sec}} = 1 \text{ Tesla.}$$



Magnetic Field: A magnetic field is a region of space where magnetic forces are exerted on charged particles in motion, arising from the movement of electric charges.

Example: The magnetic field around a bar magnet influences nearby iron filings, causing them to align along the field lines.

Magnetic Field Lines: Magnetic field lines are visual representations used to depict the direction and strength of magnetic fields, forming closed loops that indicate the path a compass needle would take.

Example: The field lines around a current-carrying wire form concentric circles, illustrating the magnetic influence of the wire's current.

FORCE ON CURRENT-CARRYING CONDUCTOR

When a current-carrying conductor is placed in a magnetic field, a force is exerted on the conductor perpendicular to both the current direction and the magnetic field direction.

Formula: The force (F) on a current (I) carrying conductor of length (L) in a magnetic field (B) is given by the formula:

$$F = I (L \times B)$$

$$F = BIL \sin \theta$$

$$B = \frac{F}{I \sin \theta}$$

TURNING EFFECT OF A CURRENT-CARRYING COIL IN A MAGNETIC FIELD

The turning effect of a current-carrying coil in a magnetic field, also known as the magnetic torque, refers to the rotational force experienced by the coil due to the interaction between its current and the magnetic field.

Formula:

The magnetic torque (τ) on a current-carrying coil with (N) turns in a magnetic field (B) is given by the formula:

$$\tau = BIAN \cos (\alpha)$$

Where:

- (I) is the current flowing through the coil,
- (A) is the area of the coil,
- (θ) is the angle between the normal to the coil's plane and the direction of the magnetic field.

FORMULA DERIVATION

Consider a rectangle coil placed in the magnetic field of strength B and the plane of the coil is parallel to the field and is free to rotate about an axis.

When current (I) passed through the coil, a force is experienced on the perpendicularly placed conductor. The magnitude of the force is $F = BIL$. Hence a pair of two equal but opposite forces (couple) acts on the coil. That causes the coil to rotate.

So,

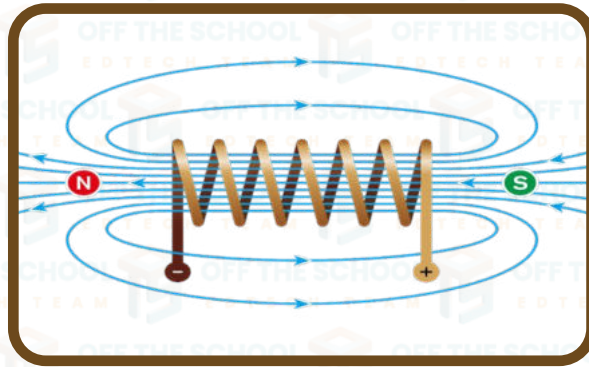
$$\text{Torque} = \tau = BIA$$

If the plane of the coil makes an angle α with the field B then the perpendicular distance $\cos(\alpha)$ can be added:

$$\tau = BIA \cos \alpha$$

If the coil has N turns, then:

$$\tau = BIAN \cos \alpha$$



DC MOTOR

A DC motor is an electric device that converts direct current (DC) electrical energy into mechanical motion by utilizing the interaction between a current-carrying coil and a magnetic field, generating a turning effect or torque that drives rotation.

TURNING EFFECT IN A DC MOTOR COIL

In a DC motor, a coil (armature) carrying current is placed within a magnetic field created by permanent magnets or electromagnets. When a current flows through the coil, it experiences a force due to the interaction with the magnetic field, following the right-hand rule. This force creates a torque (turning effect) on the coil, causing it to rotate. The direction of the current, magnetic field, and the orientation of the coil determine the direction of rotation. By reversing the current's direction, the coil's rotation can be controlled, making DC motors useful for various applications, from powering fans to driving machinery.

ELECTROMAGNETIC INDUCTION

Electromagnetic induction is the process discovered by Michael Faraday in 1831, where a changing magnetic field induces an electromotive force (EMF) or voltage in a nearby conductor, leading to the generation of an electric current.

CHANGING MAGNETIC FIELDS CAN PRODUCE INDUCE E.M.F

Changing magnetic fields induce electromotive force (EMF) in a circuit due to the fundamental relationship between magnetic flux and time-varying magnetic fields. According to Faraday's law of electromagnetic induction, when the magnetic field through a circuit changes, either by its strength varying or by the circuit's orientation changing, it creates a "magnetic push" that drives electrons to move within the circuit. This movement of electrons constitutes an electric current, generating an EMF. The key principle is that a changing magnetic field induces a voltage across a conductor, which in turn initiates the flow of electric charges, enabling the conversion of mechanical energy into electrical energy or vice versa.

FARADAY'S LAW OF INDUCTION

Faraday's law states that a changing magnetic field through a closed loop induces an electromotive force (EMF) within the loop, leading to the generation of an electric current.

FACTORS AFFECTING THE MAGNITUDE OF INDUCED EMF

Rate of Change: Faster changes in magnetic field intensity lead to greater induced EMF.

Number of Turns: More loops in the wire increase induced EMF.

Area of Loop: Larger loop area results in higher induced EMF.

Strength of Magnetic Field: Stronger magnetic fields lead to greater induced EMF.

Orientation: Changes in orientation relative to the magnetic field affect induced EMF.

LENZ'S LAW

Lenz's law is a principle in electromagnetic induction that states that the direction of an induced electromotive force (EMF) and the resulting induced current in a closed circuit will always oppose the change in magnetic field that caused them. This law ensures the conservation of energy and is a consequence of the law of conservation of energy.

Formula:

Mathematically, Lenz's law can be expressed as:

$$\text{EMF} = -N \frac{d\Phi}{dt}$$

Where (EMF) is the induced electromotive force,

$-\frac{d\Phi}{dt}$ represents the rate of change of magnetic flux through the circuit, and the negative sign signifies the opposition to the change in flux. N is the Number of turns in coil.

RELATION BETWEEN LENZ'S LAW AND CONSERVATION OF ENERGY

Lenz's Law and the conservation of energy are interconnected principles. Lenz's Law asserts that induced currents always oppose the change in magnetic flux that caused them. This opposition is crucial for preserving the energy conservation principle.

When a magnetic flux changes, an induced current emerges to counteract the change. This current generates its magnetic field, which interacts with the initial magnetic field. The work done by the induced current against the changing field converts energy into electrical energy within the circuit. This process guarantees that the energy expended to create the opposing current originates from the external source driving the change, maintaining energy conservation.

Lenz's Law safeguards energy conservation by ensuring that the energy invested in altering magnetic conditions is transferred to the induced current rather than being lost. This mechanism aligns with the core concept that energy transforms between forms, avoiding violations of energy conservation. In essence, Lenz's Law functions as a protective measure to uphold the energy conservation principle within dynamic electromagnetic systems.

A.C GENERATOR

An AC generator, also known as an alternator, is an electrical device that converts mechanical energy into alternating current (AC) electricity by rotating a coil within a magnetic field, inducing a changing magnetic flux that generates an alternating voltage.

Formula:

The formula to calculate the induced voltage (V) in an AC generator is given by

$$V = N B A \omega \sin \omega t$$

Where:

- (N) is the number of turns in the coil,
- (B) is the magnetic field strength,
- (A) is the area of the coil,
- (ω) is the angular frequency, and
- (t) represents time.

MUTUAL INDUCTION

Mutual induction is the phenomenon where a changing current in one coil induces a voltage in a neighboring coil due to the shared magnetic field, demonstrating the transfer of electromagnetic influence between the coils.

Formula:

$$E_s \propto \frac{\Delta I_p}{\Delta t}$$

$$E_s = -M \frac{\Delta I_p}{\Delta t}$$

Where M is a constant, called Mutual Inductance of the two coils.

Hence:

$$M_s = \frac{E_s}{\Delta I_p / \Delta t}$$

TRANSFORMER

A transformer is an electrical device that uses mutual induction to transfer electrical energy between two coils, typically the primary and secondary coils, by changing the voltage and current levels while maintaining the frequency.

Formula:

The basic transformer formula relates the voltages and turns of the primary (V_{primary}) and secondary ($V_{\text{secondary}}$) coils as

$$\frac{(V_{\text{secondary}})}{(V_{\text{primary}})} = \frac{(N_{\text{secondary}})}{(N_{\text{primary}})},$$

where ($N_{\text{secondary}}$) and (N_{primary}) are the number of turns in the secondary and primary coils, respectively.



APPLICATIONS

Transformers play vital roles in daily life, including:

- 1. Power Distribution:** They step up voltage for efficient long-distance transmission and step down voltage for safe local distribution in electrical grids.
- 2. Chargers:** Transformers in chargers for smartphones, laptops, and other devices convert the high-voltage AC power from outlets to lower-voltage DC suitable for charging.
- 3. Home Appliances:** Many household appliances, like refrigerators and air conditioners, use transformers to adapt voltages for their internal components.
- 4. Lighting:** Fluorescent and LED lights often have transformers to adjust voltage and current to levels suitable for lighting elements.
- 5. Audio Equipment:** Amplifiers, audio systems, and musical instruments use transformers to match impedance and isolate circuits.
- 6. Welding Machines:** Transformers provide the high currents needed for welding by stepping down the grid's high voltage.
- 7. Medical Equipment:** Medical devices require different voltage levels, and transformers help ensure safe and precise electrical supply.
- 8. Railways:** Transformers provide different voltage levels for trains, supporting their various functions and needs.